

Tyler Hatstadt

Process Essay

Perturbation-Based Simulation Modeling of Long-Latency Risk Contributions in Young-Onset Arthritis

When a process that is both highly studied and widely misunderstood stops functioning properly, I instinctively ask why. Throughout high school, this curiosity led me toward environments shaped by uncertainty and complex problem-solving. The Program for Algorithmic and Combinatorial Thinking (an extracurricular, proof-based math program) introduced me to abstraction and formal reasoning. Two years on my school's competitive physics team trained me to build quantitative models. This trajectory culminated in our first-place finish at the U.S. Invitational Young Physicists Tournament. There, I presented and defended my theoretical reasoning to a panel of expert physicists and peers. The experience reinforced a lesson that became central to my research: convincing explanations require internal coherence, empirical grounding, and resilience under critique.

Biological systems are the most intricate and unforgiving systems I have encountered. Their complexity resists tidy intuition. The messy immune system, in particular, is where this tension is most visible to me. During my junior year, this fascination inspired me to design an independent study under my AP Biology teacher, Michael Hirsch. As I explored innate and adaptive immunity, complement proteins, and T-cell functional differences, I became increasingly fascinated by the many points of failure. Because Mr. Hirsch had prior research exposure to rheumatoid arthritis, we began examining immune dysregulation. A project that started as curiosity soon pulled me into challenges that extended well beyond the biology, including several that sometimes felt more difficult to solve than the epidemiology problem itself.

Rheumatoid arthritis presented a puzzle that refused to leave me alone. The steady rise in RA incidence since 1990 in the US felt deeply illogical. Because population genotypes remain largely stable, and given the advances in disease detection and prevention alongside the decline in well-known risk factors like smoking, I initially assumed incidence might stabilize. Yet, it did not, so I began to suspect another factor must be at play. This reasoning drew my attention to obesity rates, which had risen sharply over the same period. I began to consider a possibility: what if obesity were contributing to RA pathogenesis rather than merely coinciding with it?

Before I could dive into my hypothesis, I hit the end of the school year, which meant losing the steady guidance from Mr. Hirsch. Around the same time, a planned internship with a professor of biostatistics at the Yale School of Public Health fell through, removing what I had hoped would become a critical source of mentorship. Because of earlier academic commitments, I had missed the formal research track at my school, which required me to create my own research environment. I attempted to continue developing my ideas independently by building regression models to surface patterns. But this went nowhere. Datasets of young RA patients are limited and private. Also, at the time, I assumed such data would be out of reach for an independent high school researcher. I found myself stuck without a clear path forward.

Later that summer, I met Michael Bryan through Inspirit AI, who became my research mentor. He is an MD/PhD student at the University of Oxford whose research focuses on immuno-oncology, and he helped breathe new energy into my project. From our very first Zoom call, he helped me understand how to seriously engage in statistical and epidemiological research. Through our sessions, I realized early on that waiting for the ‘perfect dataset’ was itself a methodological trap. If the exposure occurred decades before diagnosis, no existing cohort could cleanly capture it. And it was that realization that forced me to pivot toward simulation, not as a shortcut, but as the only practical way to test counterfactual scenarios. To manage the project, I divided it into phases: literature synthesis, model construction, validation and sensitivity analyses, and manuscript preparation.

I developed my model in Java, building a Monte Carlo simulation designed to generate a synthetic database of one million individuals, large enough to ensure stable population-level prevalence estimates. To reflect real population variability, the model assigned probabilistic risk factor profiles. These were based on literature-derived odds ratios, baseline prevalence estimates, and demographic distributions, enabling us to compute prevalence patterns across age groups. Consistent with our hypothesis, I implemented an age-dependent exponential decay function to model attenuation of BMI-related risk and ran multiple sensitivity analyses to see whether there were any significant deviations in results, and observed none. To ensure the simulation produced clinically realistic estimates, we validated predicted RA prevalence against Global Burden of Disease benchmarks and systematically perturbed BMI across populations, observing counterfactual scenarios not easily sampled in traditional epidemiological studies. Uncertainty of population-level statistics was quantified using a combination of confidence-interval distributions for input parameters and a bootstrap resampling method over 2,500 iterations.

The model did not cooperate politely at first. Early iterations failed outright, producing outputs that were implausible, the data were incomprehensible, and the code was messy and difficult to debug. This made even small corrections painfully inefficient. Even so, because this work was not conducted for a grade in school, I was able to experiment more freely without fear of penalty, and that freedom proved essential. In hindsight, many of the project’s most important advances emerged directly from failures that exposed flawed assumptions, unstable parameters, and conceptual gaps in the model.

While working on this project at home over the summer, I chose to rebuild the simulation multiple times to create a structure that was cleaner, more modular, and easier to modify, rather than patching isolated errors. Through discussions with Michael and repeated revisions, I refined both the computational framework and its embedded assumptions. Running high-performance simulations in Java and analyzing and visualizing outputs in Python allowed me to gradually constrain a plausible range for BMI’s population-level contribution to RA prevalence. Once the outputs stabilized within clinically realistic bounds, I shifted from large structural changes to targeted calibrations to tighten estimates and improve the accuracy of the model’s confidence intervals. This process lasted several months, with each iteration proving increasingly satisfying as my models finally came to life.

Developing this model forced me to confront a broader methodological issue. Traditional observational epidemiology is often poorly suited for studying long-latency exposures, where early exposures may precede diagnosis by decades. Even when associations exist, small effects at the individual level can accumulate into

meaningful shifts at the population scale. Rather than attempting to claim causality, my model was designed to test plausibility: whether known risk relationships, when perturbed across realistic obesity scenarios, could reproduce observed prevalence patterns. Simulation became not just a computational tool, but a framework for exploring counterfactuals that observational data cannot directly address.

Excited by my progress, I felt ready to test my work against the outside world. I wrote an abstract and began drafting a manuscript. The timing was off-cycle for high school science fairs, so I reached out to the American Association of Immunologists to request consideration as a poster presenter at their Midwinter Conference. After an independent review, I received formal acceptance to present, where I was the only high school student presenter.

Transitioning from a long-form manuscript to a poster required severe compression, which was a valuable experience. Condensing the research into its core components helped my conclusions crystallize, assumptions clarify, and next steps become evident. Turning complex modeling decisions into diagrams, concise explanations, and defensible claims sharpened my thinking and began to bring the framework's significance into sharper focus. I traveled to California with my mother for the conference, while missing school and athletic commitments. What initially felt like a logistical challenge became intellectually illuminating.

Immersing myself for four days in the broader scientific community was transformative. I received meaningful feedback from clinical researchers, including a few from the Allen Institute. I also connected with faculty interested in extending the model to other diseases and gained perspective on how my work fits within the research landscape.

Strongly positive feedback from the conference strengthened my confidence and refined the next steps for my work. I posted my manuscript to bioRxiv. Contemporaneously with my poster work, I had been contacting clinical researchers to explore potential collaborations involving their de-identified patient data. After numerous proposals and rejections, I connected with two groups that agreed: Seattle Children's Hospital and Research Center, and a group at King's College London with access to the ERAS-ERAN Steering Group dataset. After multiple calls, emails, and proposal revisions, we reached agreements in principle and are awaiting completion of the data transfer agreements. Pending analysis of the new data, our final stage will be to update the manuscript to submit for first-author publication in a peer-reviewed immunology journal.

After nearly fourteen months of sustained work, this project fundamentally changed my understanding of both research and my own thinking. I began with a question that felt logically inconsistent, assuming the primary challenge would be analytical. Instead, I encountered failed models, unstable assumptions, lost opportunities, data access barriers, and long periods of uncertainty. The experience taught me that meaningful research is rarely a linear progression of correct ideas, but an iterative process shaped by error, revision, and persistence. I learned to treat failure not as a verdict, but as information; a mechanism that exposes hidden weaknesses and forces clearer thinking. Most importantly, I discovered that my drive to solve complex problems does not depend on grades, deadlines, or guaranteed outcomes. It is driven by the simple, stubborn desire to understand systems that resist easy explanation.